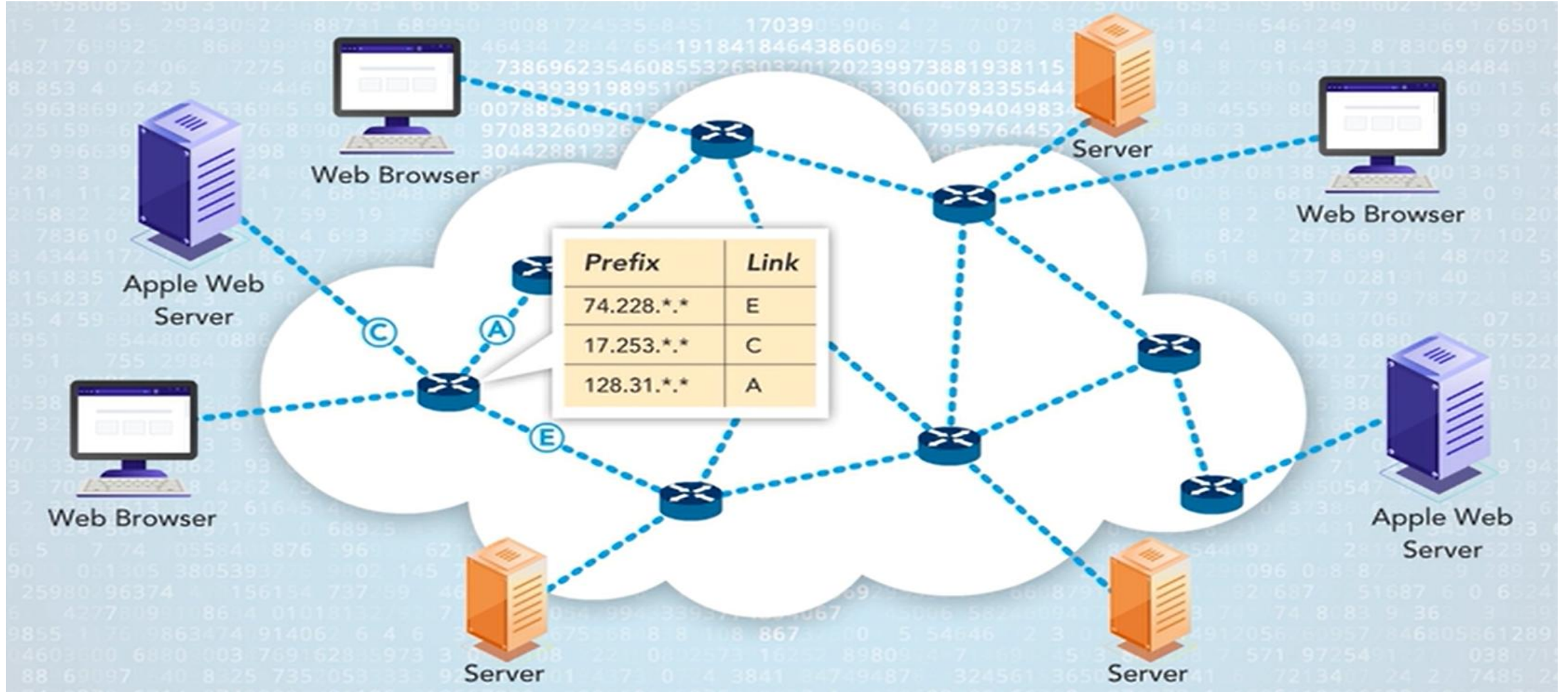




IPv4 Addressing & Network Layer

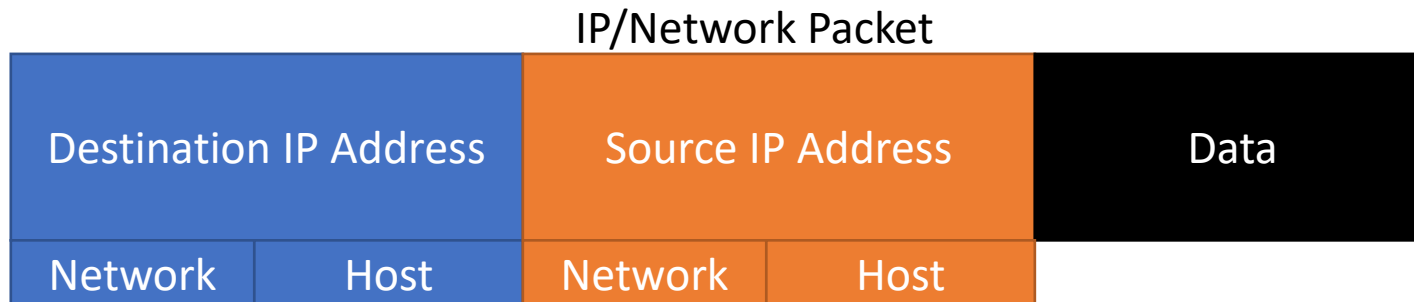
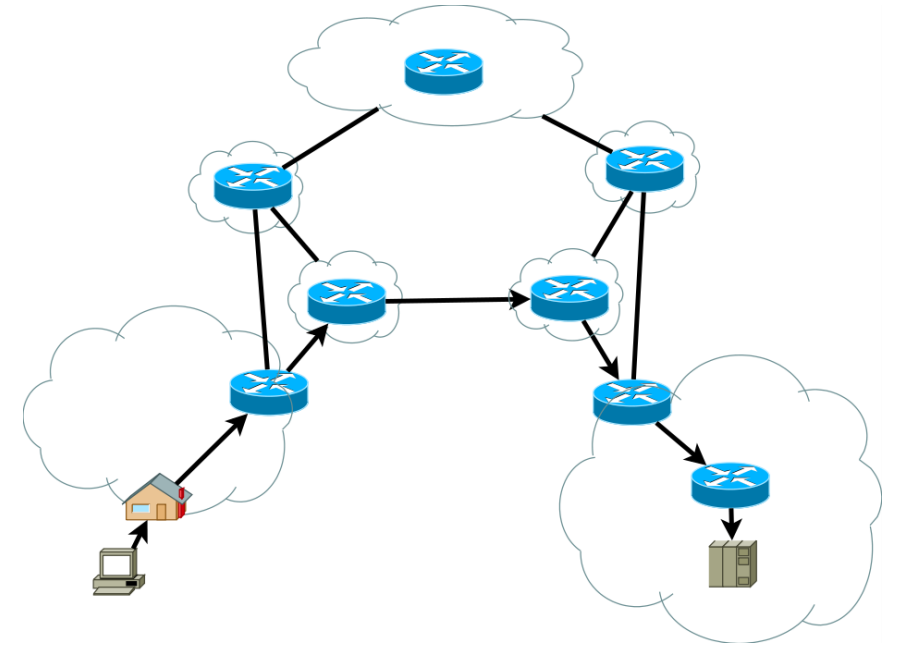
By: Mujahid Tabassum - 2025

Agenda

- Network layer
- IP Characteristics
- IP Addressing
- IP Network Mask
- Introduction to Routing
 - Host Routing
 - Router Routing

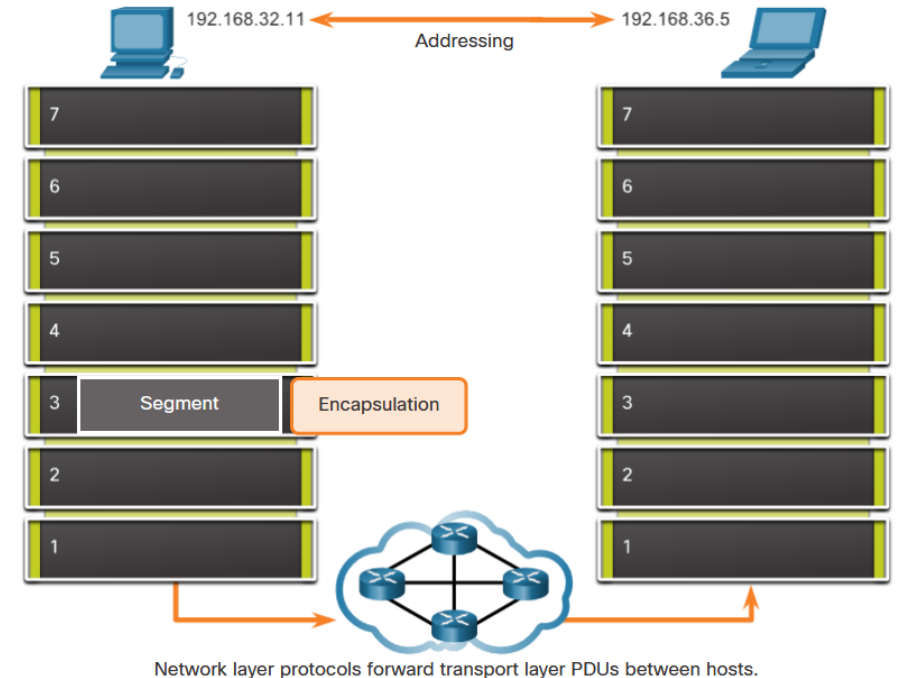
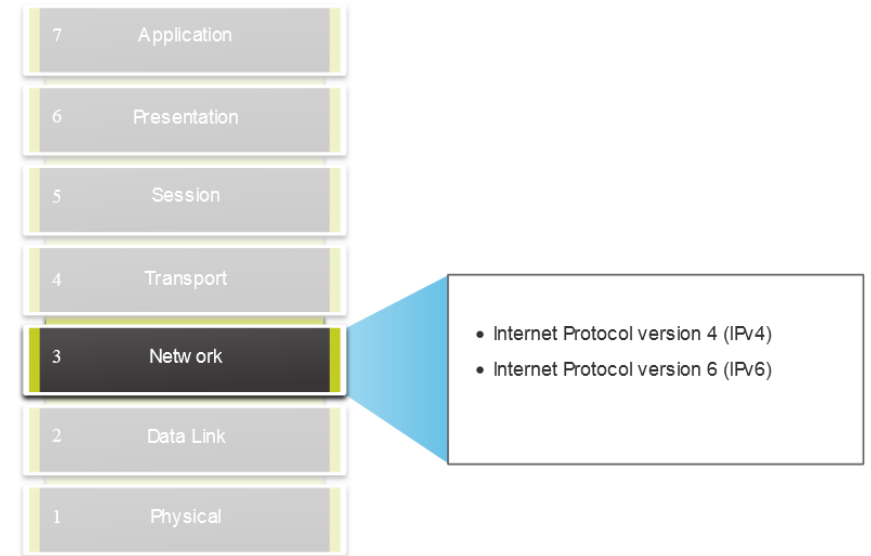
Recap: Internet Layer Communication

- Internet/Network layer protocols primary function is to move data from one network to another network
- Network addresses(IP Addresses) must have a mechanism to locate hosts on different networks
- Routers use the network portion of this address to determine which path to use to reach its destination.



The Network Layer

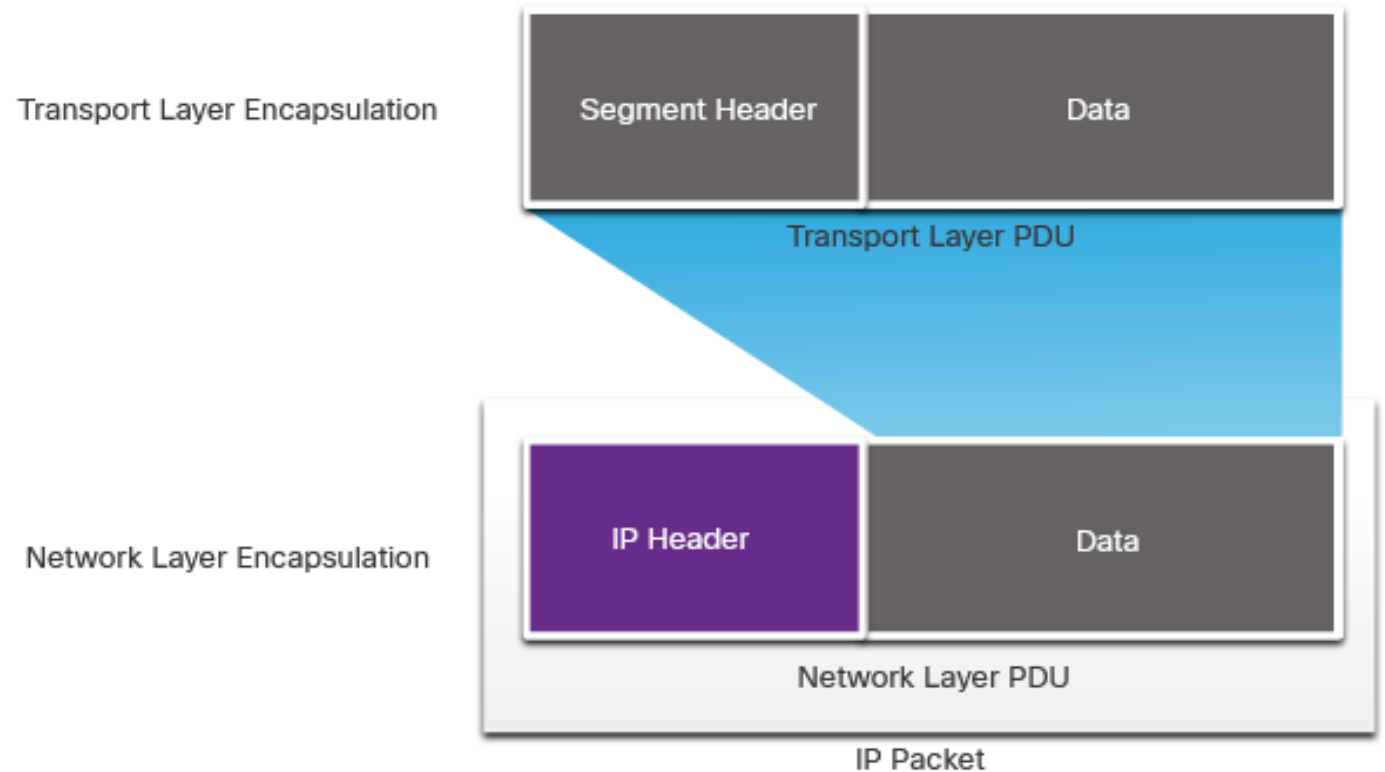
- Provides services to allow end devices to exchange data
- IP version 4 (IPv4) and IP version 6 (IPv6) are the principle of network layer communication protocols.
- The network layer performs four basic operations:
 - Addressing end devices
 - Encapsulation
 - Routing
 - De-encapsulation



IP Encapsulation

- IP encapsulates the transport layer segment.
- IP can use either an IPv4 or IPv6 packet and not impact the layer 4 segment.
- IP packet will be examined by all layer 3 devices as it traverses the network.
- The IP addressing does not change from source to destination.

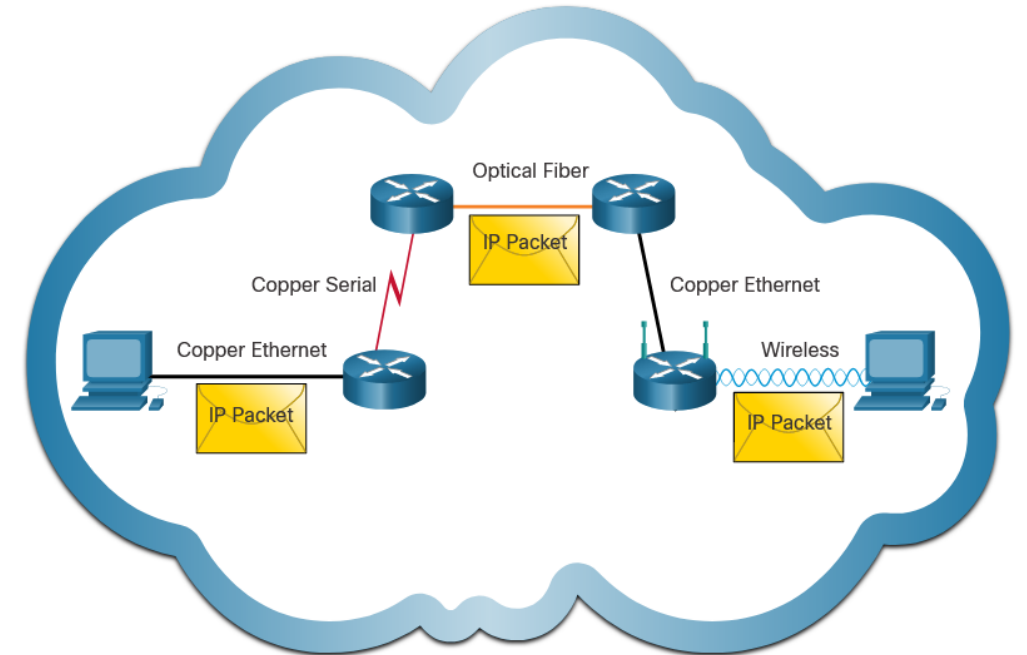
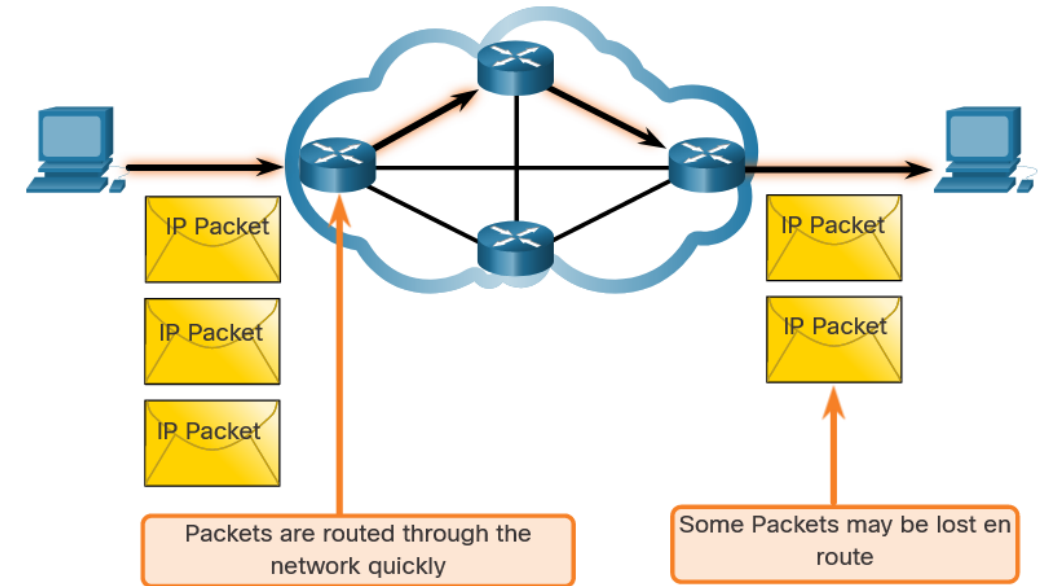
Note: NAT will change addressing, but will be discussed in a later module.



Characteristics of IP

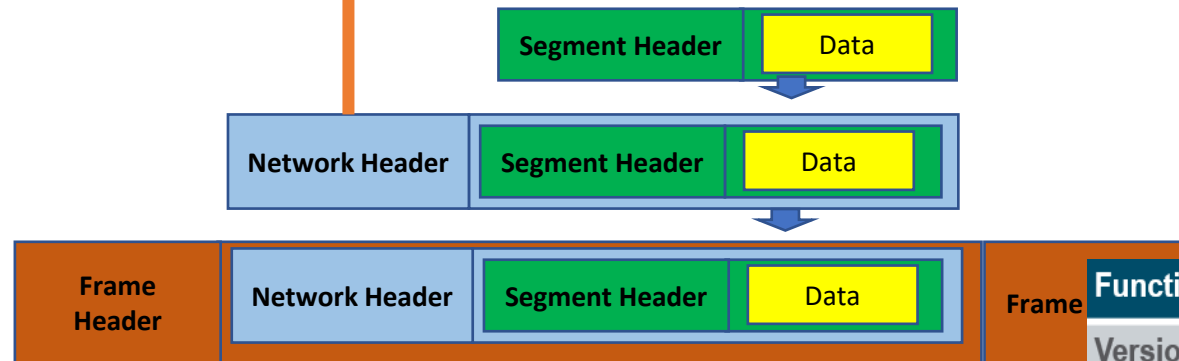
IP is meant to have low overhead and may be described as:

- Connectionless
- Best Effort
- Media Independent



The IP Packet Header

0								1								2								3							
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31
Version				IHL				DSCP						ECN		Total Length															
Identification																Flags			Fragment Offset												
Time To Live								Protocol								Header Checksum															
Source IP Address																															
Destination IP Address																															
Options(if any)																															



Function	Description
Version	This will be for v4, as opposed to v6, a 4 bit field= 0100
Differentiated Services	Used for <u>QoS</u> : <u>DiffServ</u> – DS field or the older <u>IntServ</u> – <u>ToS</u> or Type of Service
Header Checksum	Detect corruption in the IPv4 header
Time to Live (TTL)	Layer 3 hop count. When it becomes zero the router will discard the packet.
Protocol	I.D.s next level protocol: ICMP, TCP, UDP, etc.
Source IPv4 Address	32 bit source address
Destination IPV4 Address	32 bit destination address

Aside: IPv4 Limitations

- Running out of IPv4 addresses
- Because of this, lack of end to end connectivity
 - Overcome using private IP addressing and Network Address Translation
- NAT is everywhere, but was only intended as a temporary solution
- Solution: IPv6
 - **Increased address space** – based on 128 bit address, not 32 bits
 - **Improved packet handling** – simplified header with fewer fields
 - **Eliminates the need for NAT**
- **IPv4 remains dominant choice for various reasons...**

An IPv6 address (in hexadecimal)

2001:0DB8:AC10:FE01:0000:0000:0000:0000



2001:0DB8:AC10:FE01::

Zeros can be omitted

A diagram showing the binary expansion of the IPv6 address segments. Four curved arrows point from the first four segments of the compressed address (2001, 0DB8, AC10, FE01) to the first four segments of a long binary string. The binary string is divided into two lines. The first line contains the binary for the first four segments: 0010000000000001:0000110110111000:1010110000010000:111111000000001:. The second line contains the binary for the remaining four segments, all of which are zero-filled: 0000000000000000:0000000000000000:0000000000000000:0000000000000000.

0010000000000001:0000110110111000:1010110000010000:111111000000001:
0000000000000000:0000000000000000:0000000000000000:0000000000000000

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IPv6 Overview

- IPv6 was developed by Internet Engineering Task Force (IETF).
- IPv6 overcomes the limitations of IPv4.
- Improvements that IPv6 provides:
 - **Increased address space** – based on 128-bit address, not 32 bits
 - **Improved packet handling** – simplified header with fewer fields
 - **Eliminates the need for NAT** – since there is a huge amount of addressing, there is no need to use private addressing internally and be mapped to a shared public address

IPv4 and IPv6 Address Space Comparison

Number Name	Scientific Notation	Number of Zeros
1 Thousand	10^3	1,000
1 Million	10^6	1,000,000
1 Billion	10^9	1,000,000,000
1 Trillion	10^{12}	1,000,000,000,000
1 Quadrillion	10^{15}	1,000,000,000,000,000
1 Quintillion	10^{18}	1,000,000,000,000,000,000
1 Sextillion	10^{21}	1,000,000,000,000,000,000,000
1 Septillion	10^{24}	1,000,000,000,000,000,000,000,000
1 Octillion	10^{27}	1,000,000,000,000,000,000,000,000,000
1 Nonillion	10^{30}	1,000,000,000,000,000,000,000,000,000,000
1 Decillion	10^{33}	1,000,000,000,000,000,000,000,000,000,000,000
1 Undecillion	10^{36}	1,000,000,000,000,000,000,000,000,000,000,000,000

Legend



There are 4 billion IPv4 addresses

There are 340 undecillion IPv6 addresses

IPv4 Addressing

IPv4 Address: Phone number analogy

- Phone numbers are like network addresses

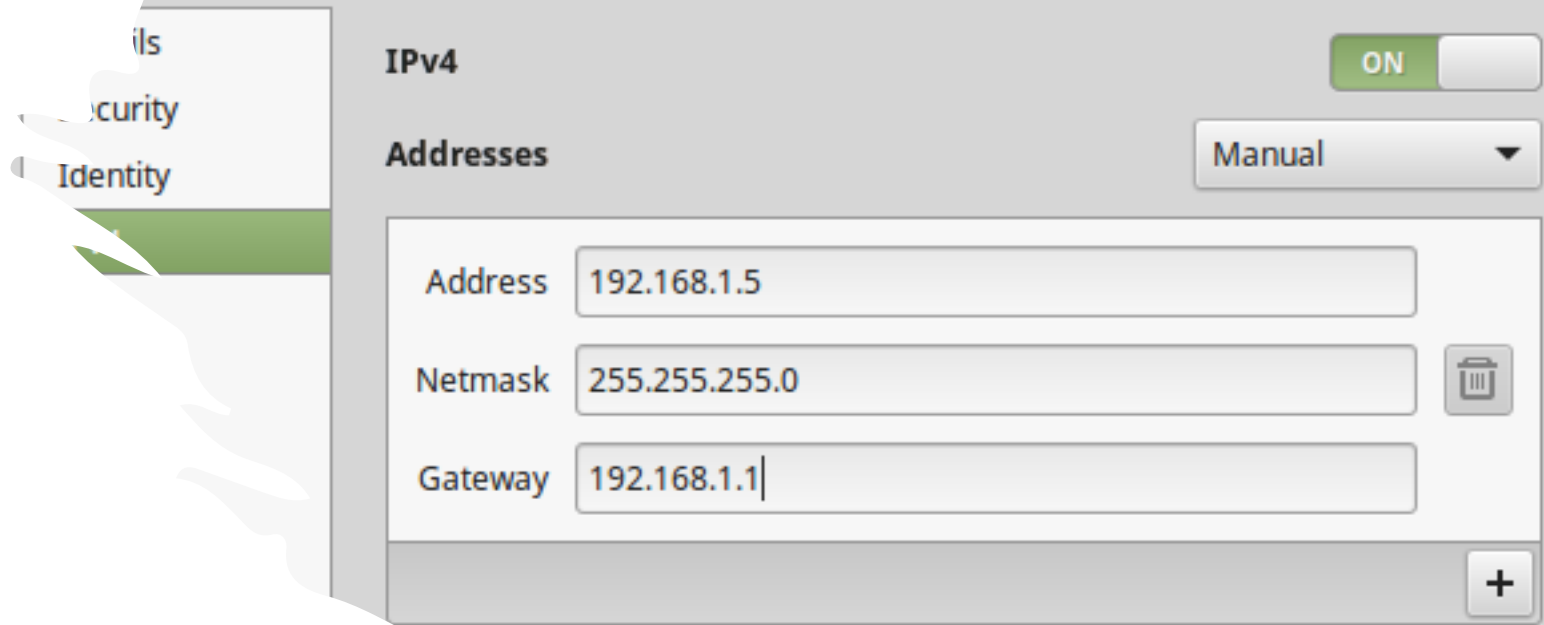
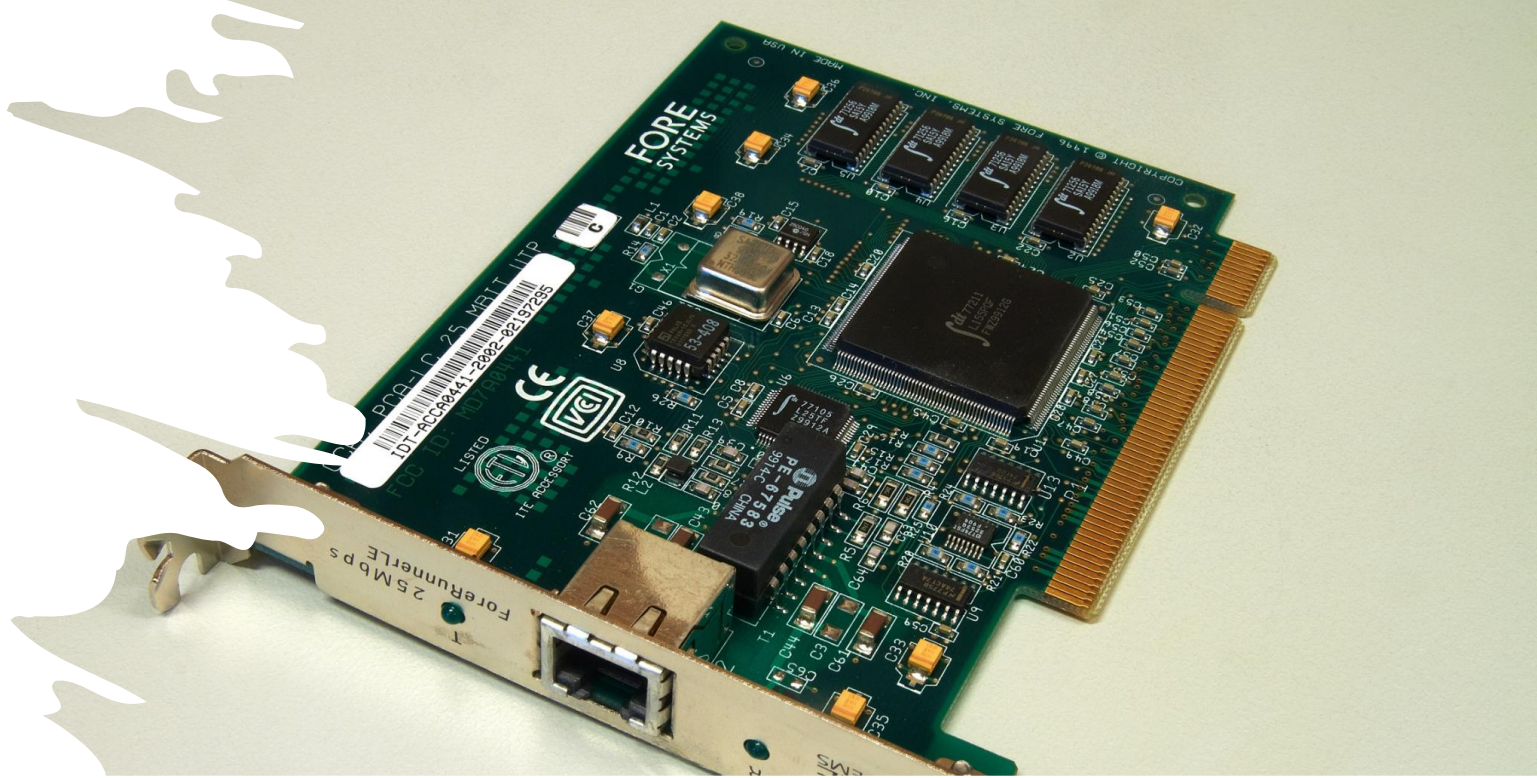
Area Code – similar
to network portion

Phone number –
similar to host portion

051-3025635

MAC and IP Address

- Physical/MAC address
 - Flat Addressing Scheme
 - "Burned" into Network Interface
 - Doesn't change (like a PPS number)
- IP Address
 - Hierarchical Addressing
 - Software defined
 - Your devices "mailing" address (ie it usually changes when you move)
- You need both...



MAC is flat addressing

- If the Internet was a flat network with only MAC addresses, switches would need to know the **billions** of MAC addresses or broadcast the frame as an unknown unicast.



IP is Hierarchical

- IP address indicates what network the packet belongs to.
- Routers maintain lists of IP network addresses to route the packet to the right network.

SETU



UCD

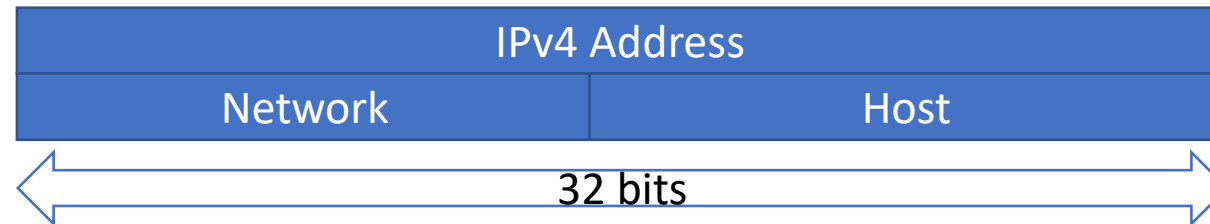


Your ISP



IPv4 Addressing

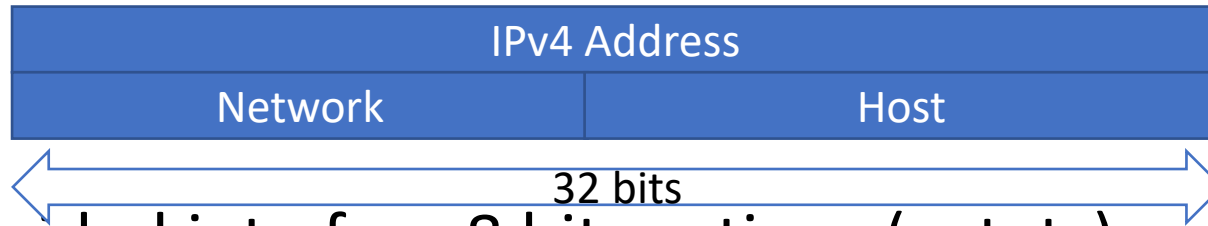
- Which bits refer to network number???



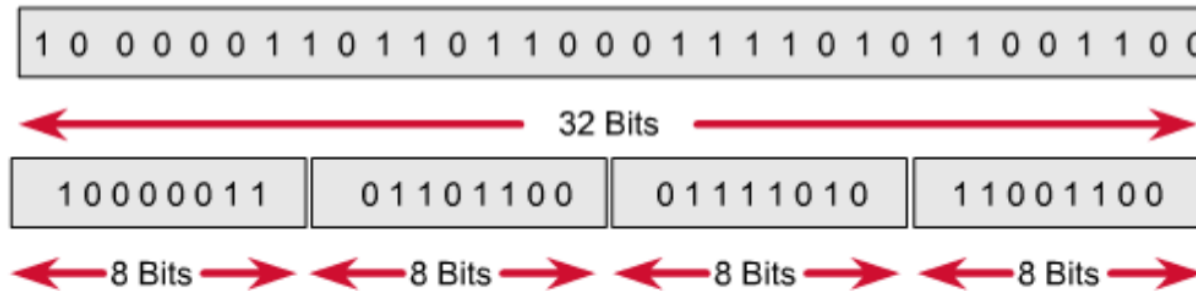
- In the past – Classful IP addressing
 - Value of first octet(8 bits) determined the network/host portion
- Currently
 - Classless IP addressing
 - Use a **subnet mask** to determine network portion of address
- Classless IP Addressing is what is used within the Internet and in most internal networks.

IPv4 Addressing

- IPv4 are 32 bits



- Divided into four 8 bit sections (octets)



- Typically expressed in **Dot Decimal Notation**



IPv4 Address: Converting from binary to dot decimal notation

1 0 0 0 0 0 1 1 0 1 1 0 1 1 0 0 0 1 1 1 1 0 1 0 1 1 0 0 1 1 0 0

32 Bits

10000011 01101100 01111010 11001100

8 Bits 8 Bits 8 Bits 8 Bits

128	64	32	16	8	4	2	1
1	0	0	0	0	0	1	1

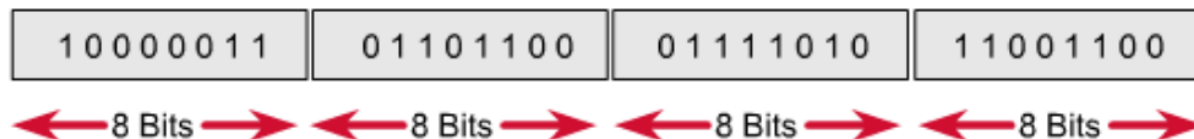
128 + 0 + 0 + 0 + 0 + 0 + 2 + 1

131.

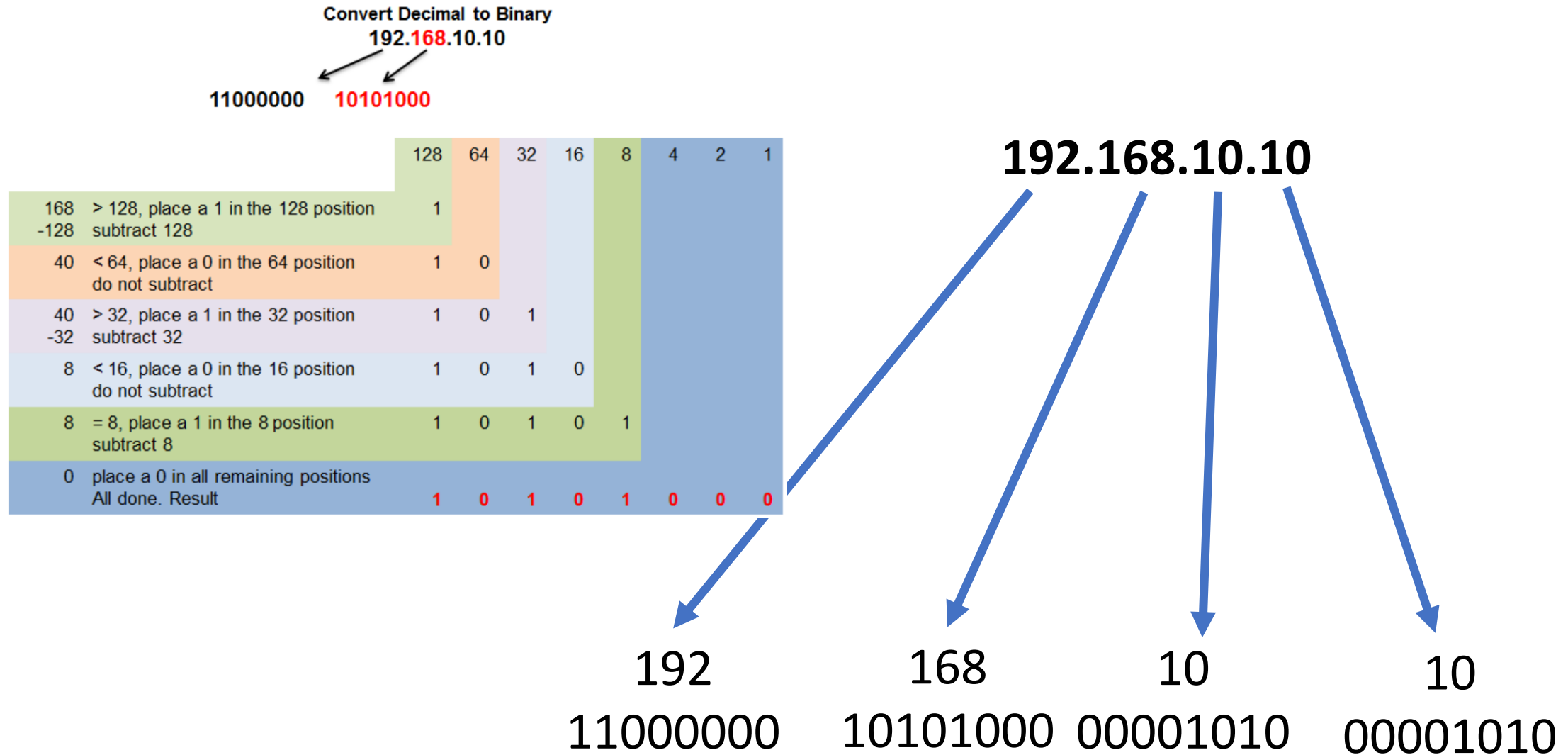
IPv4 Address: Converting from Decimal to Binary



128	64	32	16	8	4	2	1



IPv4 Address: Converting Decimal to Binary



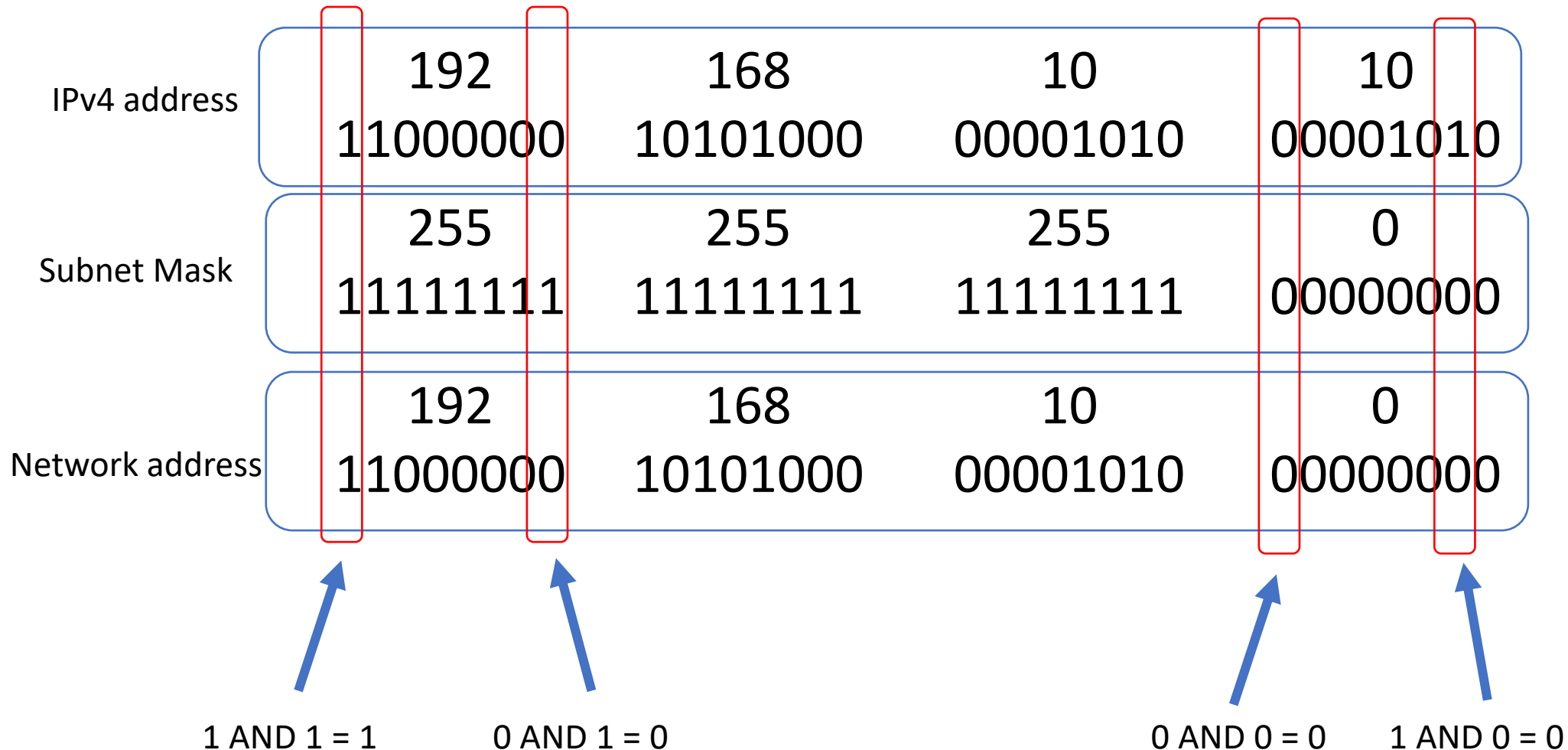
IPv4 Subnet Mask

- A subnet mask is a separate 32-bit pattern used to define the network and host portions of an address.
- Shows where the network portion is in the IP.

IPv4 address	192 11000000	168 10101000	10 00001010	10 00001010
Subnet Mask	255 11111111	255 11111111	255 11111111	0 00000000
Network address	192 11000000	168 10101000	10 00001010	0 00000000

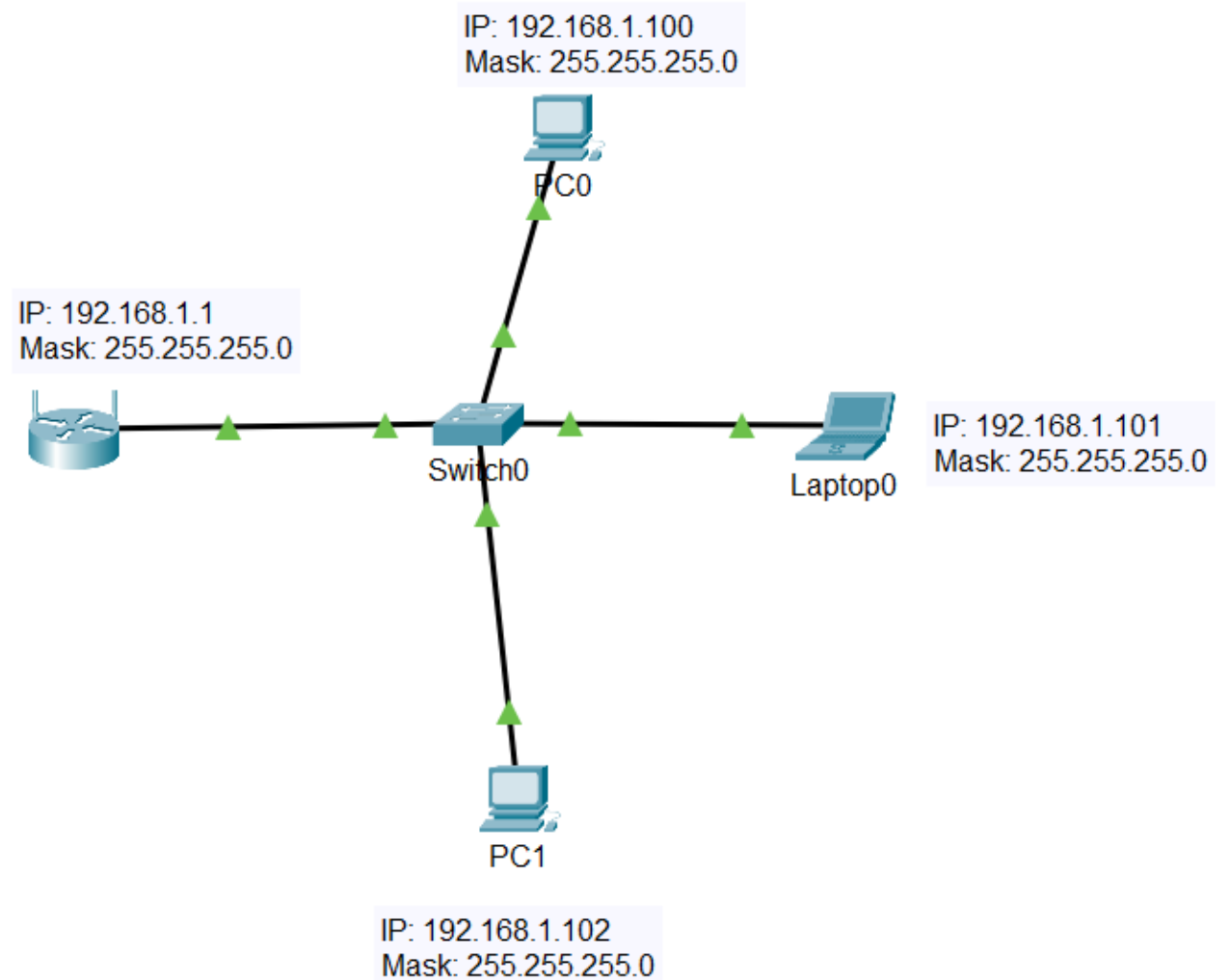
Subnet mask always has 1s from left to final position of network address. 1s show the location of network portion

IPv4 Address: Subnet Mask – bitwise AND operation



IP Addressing: Network, Host, Broadcast

- **Network address** (host portion all 0s) – Network
 - In opposite example, 192.168.1.0
- **Broadcast address** (host portion all 1s)
 - In opposite example, 192.168.1.255
- **First host address**
 - 192.168.1.1
- **Last host address**
 - 192.168.1.254



IP Addressing: Network Address Prefix Length

- Alternative way to identify subnet mask.
- Known as “slash notation”
- Example:

IP Network Address 10.1.0.0/23

/23 indicates number of bits
"borrowed" for network address

Subnet mask is : 255.255.254.0 (11111111.11111111.11111110.00000000)

10.1.0.0 00001010.00000001.00000000.00000000 (network address)

10.1.0.1 00001010.00000001.00000000.00000001 (first host)

10.1.1.254 00001010.00000001.00000000.11111110 (last host)

10.1.1.255 00001010.00000001.00000000.11111111 (broadcast)

Max Number of hosts: $2^9 - 2 = 510$ hosts

Max number of hosts = $2^b - 2$
where b is the number of bits used for
the HOST portion of the IP Address.
(take away 2 as you **CANNOT** use
network or broadcast address for
hosts.)

First and Last Host Address

Network Address 10.1.0.0/22 00001010.00000001.00000000.00000000
255.255.252.0 11111111.11111111.11111100.00000000

10.1.0.1 00001010.00000001.00000000.00000001 (first)
10.1.3.254 00001010.00000001.00000011.11111110 (last)
10.1.3.255 00001010.00000001.00000011.11111111 (broadcast)

Number of hosts: $2^{10} - 2 = 1024 - 2 = 1022$ hosts

IPv4 Dot Decimal Prefix Notation

	<u>Dotted Decimal</u>	<u>Significant bits shown in binary</u>
Network Address	10.1.1.0/24	10.1.1.00000000
First Host Address	10.1.1.1	10.1.1.00000001
Last Host Address	10.1.1.254	10.1.1.11111110
Broadcast Address	10.1.1.255	10.1.1.11111111
Number of hosts: $2^8 - 2 = 254$ hosts		

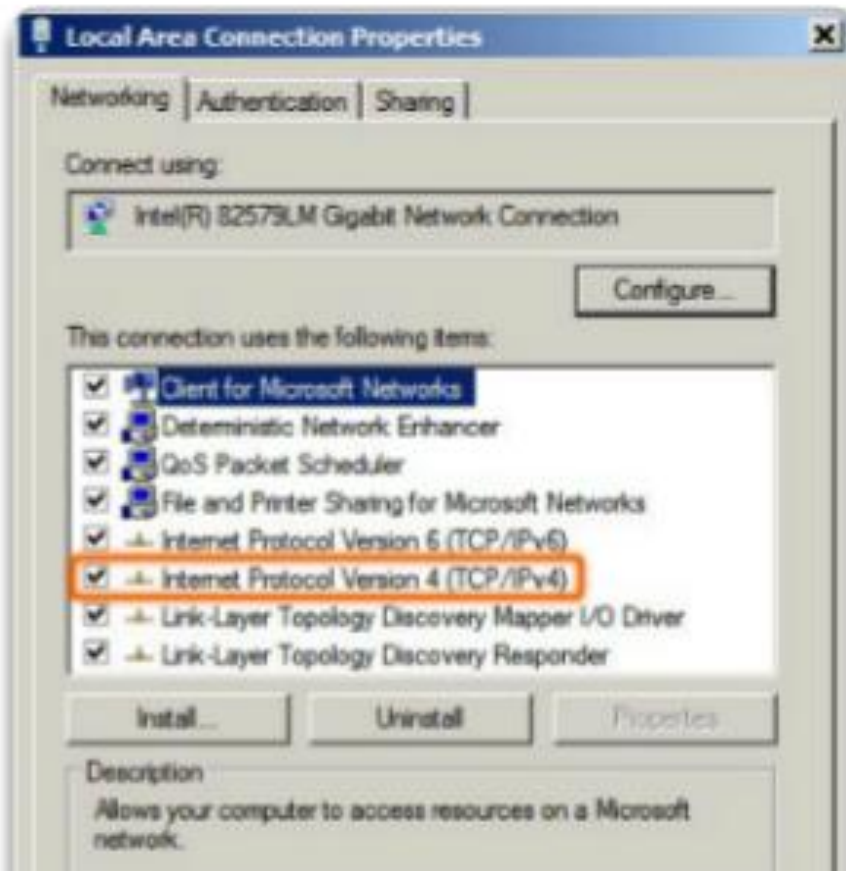
Network Address	10.1.1.0/25	10.1.1.00000000
First Host Address	10.1.1.1	10.1.1.00000001
Last Host Address	10.1.1.126	10.1.1.01111110
Broadcast Address	10.1.1.127	10.1.1.01111111
Number of hosts: $2^7 - 2 = 126$ hosts		

Network Address	10.1.1.0/26	10.1.1.00000000
First Host Address	10.1.1.1	10.1.1.00000001
Last Host Address	10.1.1.62	10.1.1.00111110
Broadcast Address	10.1.1.63	10.1.1.00111111
Number of hosts: $2^6 - 2 = 62$ hosts		

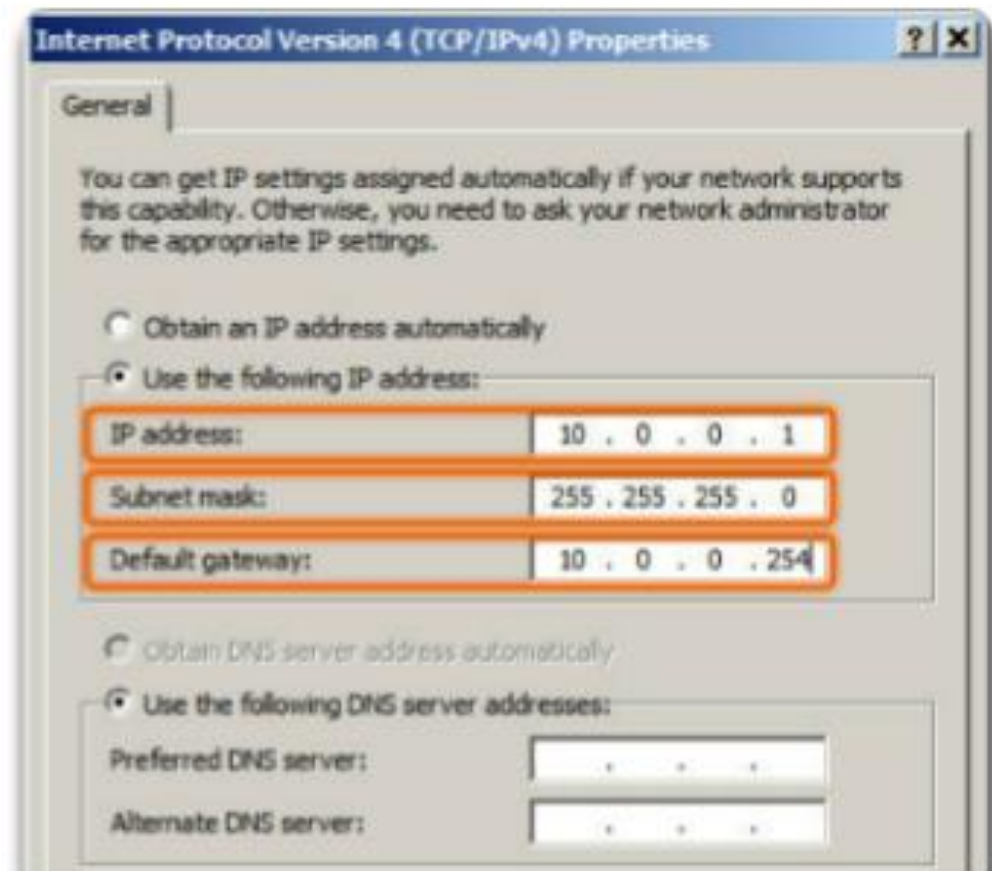
Assigning Static IP to a Host

- Windows example

LAN Interface Properties



Configuring a Static IPv4 Address



Assigning Static IP to a Host

- Packet Tracer

File Server

Physical Config Services **Desktop** Programming

IP Configuration

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address

Subnet Mask

Default Gateway

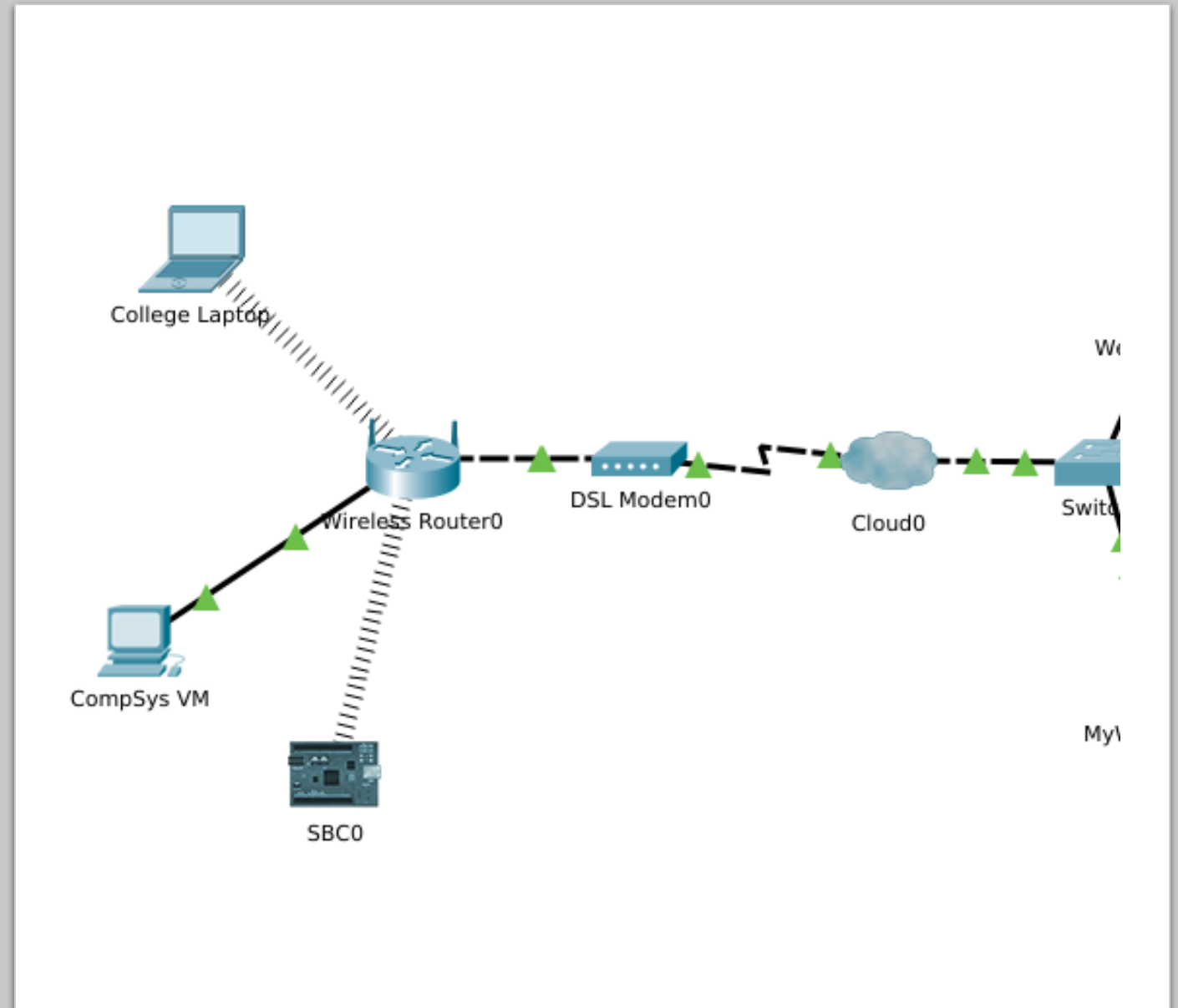
DNS Server

Routing on a Host

e.g. your computer

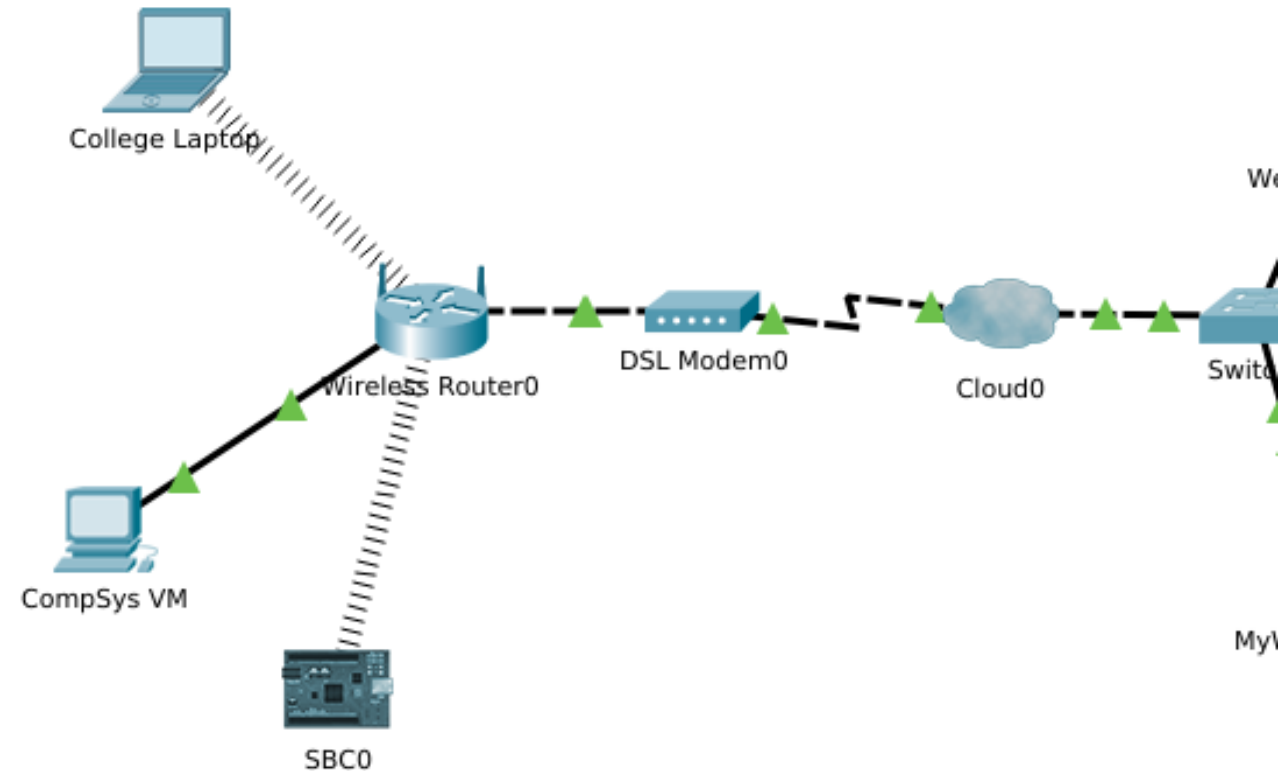
Forwarding:
where should it
go?

- All hosts maintain a **Routing Table**
- The host will send an IP packet to
 - Itself – loopback address(127.0.0.1)
 - Local Hosts – Same LAN (part of IP) matching network
 - Remote Host – On different LAN(different network part of IP)



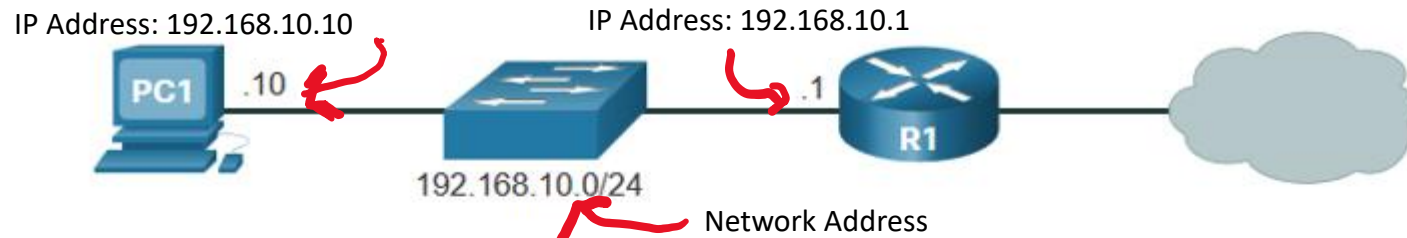
Forwarding: where should it go?

- The Source device determines whether the destination is local or remote using it's own IP address and subnet mask and destination IP address
- Local destination traffic is sent straight out from interface and is handled by Data-Link/layer 2 devices
- Remote traffic is forwarded to **Default Gateway**
- The **Default Gateway** will:
 - Have an IP address on LAN
 - Accept and route to other networks.
- Last Resort Route: if no other route matches



Host Routing Table

- **route print, route** or **netstat -r** will display the host routing table



IPv4 Routing Table for PC1

Default Gateway

```
C:\Users\PC1> netstat -r
```

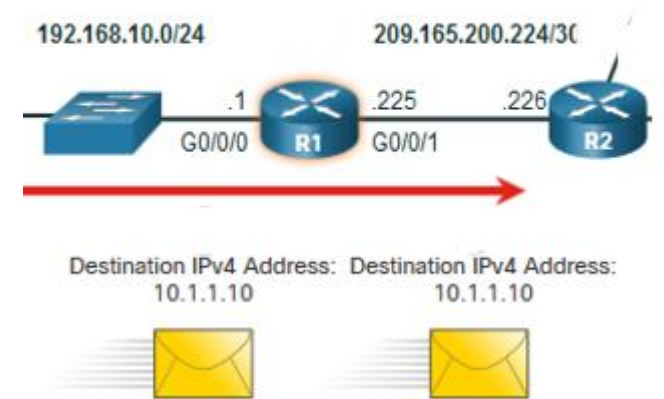
```
IPv4 Route Table
```

```
Active Routes:
```

Network	Destination	Netmask	Gateway	Interface	Metric
	0.0.0.0	0.0.0.0	192.168.10.1	192.168.10.10	25
	127.0.0.0	255.0.0.0	On-link	127.0.0.1	306
	127.0.0.1	255.255.255.255	On-link	127.0.0.1	306
	127.255.255.255	255.255.255.255	On-link	127.0.0.1	306
	192.168.10.0	255.255.255.0	On-link	192.168.10.10	281
	192.168.10.10	255.255.255.255	On-link	192.168.10.10	281
	192.168.10.255	255.255.255.255	On-link	192.168.10.10	281
	224.0.0.0	240.0.0.0	On-link	127.0.0.1	306
	224.0.0.0	240.0.0.0	On-link	192.168.10.10	281
	255.255.255.255	255.255.255.255	On-link	127.0.0.1	306
	255.255.255.255	255.255.255.255	On-link	192.168.10.10	281

Routers – Packet Forwarding

- On receiving frame from host device
 1. Decapsulates/extracts IP Packet from Ethernet Frame
 2. Gets IP destination address
 3. Searches Routing table for best match and finds “next hop”
 4. Encapsulates IP packet back into Ethernet/Data Link Frame
 5. Forwards on to next hop device

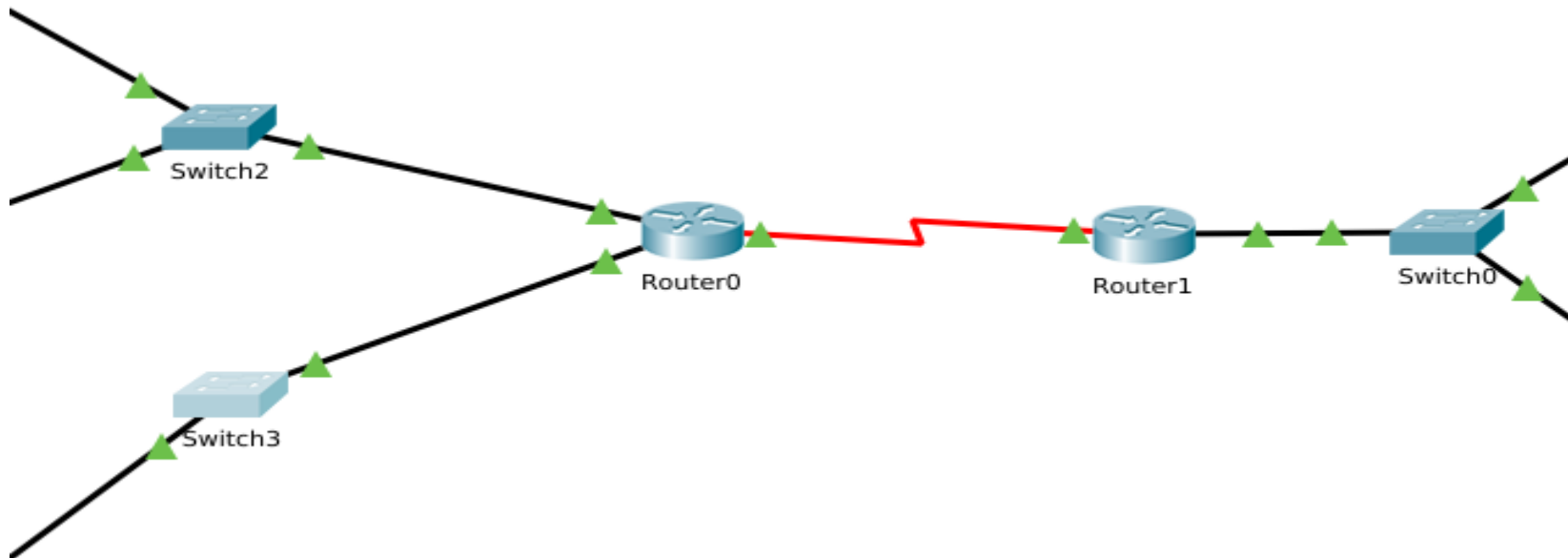


R1 Routing Table

Route	Next Hop or Exit Interface
192.168.10.0 /24	G0/0/0
209.165.200.224/30	G0/0/1
10.1.1.0/24	via R2
Default Route 0.0.0.0/0	via R2

IP Routing Table

- Routing tables can contain:
 - Directly Connected routes: active interfaces automatically added by router
 - Remote: Not directly connected but entry made either manually (static route) or Dynamically(routing protocol)
 - Default Route: where to send traffic if no other match found.



Static/Dynamic Routing

- Static routes are configured manually
- Must be Updated if topology changes
- Fine for small, non- redundant networks
- Combined with routing protocols for defining default routes.
- Dynamic Routing can:
 - Discover remote networks
 - Exchange routing info with other routers
 - Choose the best path to the destination
 - Handle topology changes, dropped links.

IP Addressing: Types

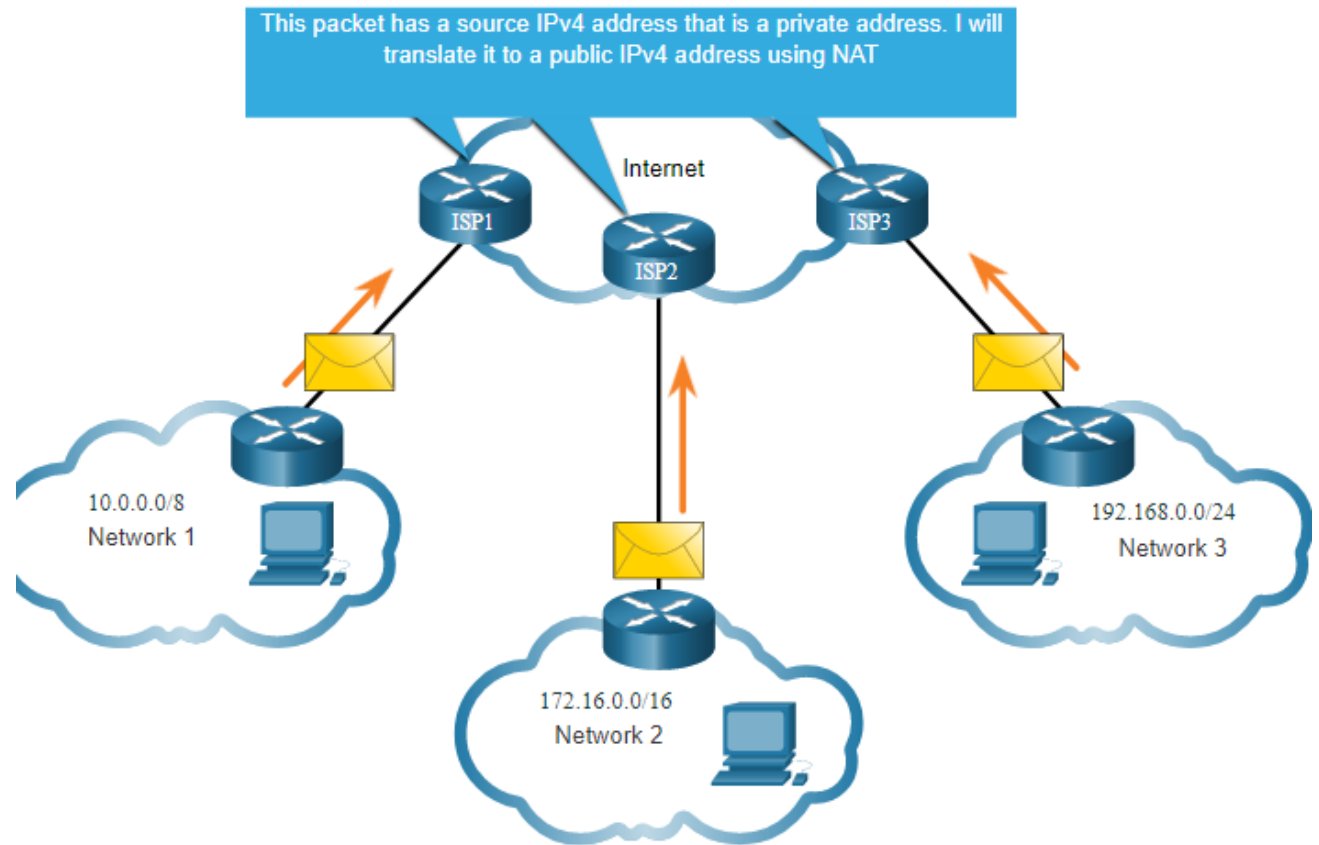
IP Addressing: Address Types

- Public IPv4 addresses are globally routed between internet service provider (ISP) routers.
- Private addresses are common blocks of addresses used by most organisations to assign IPv4 addresses to internal hosts.
 - Not unique
 - Not routed globally

Network	Range (RFC 1918)
10.0.0.0/8	10.0.0.0-10.255.255.255
172.2.16.0.0/12	172.16.0.0 – 172.31.255.255
192.168.0.0/16	192.168.0.0-192.168.255.255

IP Addressing: NAT

- Network Address Translation (NAT) translates private IPv4 addresses to public IPv4 addresses.
- NAT usually on router connected to internet(it's on your home router)
- Translates between private and public global IP addresses.



IP Addressing: History

- RFC 790 (1981) allocated IPv4 addresses in classes
 - But no medium size networks
- Class A/B addresses depleted quickly
- Lots of wasted addresses
- REPLACED BY CLASSLESS ADDRESSING

Class A	Network	Host		
Octet	1	2	3	4

Class B	Network		Host	
Octet	1	2	3	4

Class C	Network			Host
Octet	1	2	3	4

Class D	Host			
Octet	1	2	3	4

Address Class	First Octet Range	Number of Possible Networks	Number of Hosts per Network
Class A	0 to 127	128 (2 are reserved)	16,777,214
Class B	128 to 191	16,348	65,534
Class C	192 to 223	2,097,152	254

IP Address Assignment

- IANA (Internet Assigned Numbers Authority)responsible for global coordination of the Internet Protocol addressing systems. Regional-based and hierarchical...

